A finite-lattice Lorentz-violation bound from interval-arithmetic enclosure of J_1 on a shell-adaptive BCC lattice at $N = 256$: $J_1 \in [5.99 \times 10^{-2}, 1.51 \times 10^{-1}]$ excludes $J_1 = 0$

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We compute a finite-lattice deterministic-enclosure bound on the Lorentz-violation parameter J_1 (deviation of the lattice light-cone structure from naive Lorentz invariance) on a shell-adaptive BCC lattice with $N = 256$ sites per dimension within the Topological Energy Condensate Theory (TECT). Using rigorous interval arithmetic (`mpmath` library, arbitrary precision) to account for all rounding and truncation errors, the deterministic enclosure of J_1 is

$$J_1 \in \left[\, 5.99 \times 10^{-2},\ 1.51 \times 10^{-1} \,\right].$$

The interval explicitly EXCLUDES the violation-free value $J_1 = 0$, so this is a finite-lattice Lorentz-VIOLATION bound, NOT a Lorentz-invariance certification. The enclosure is a *deterministic* interval (in the interval-arithmetic sense), not a 95% probabilistic confidence interval. The bound is consistent with effective Lorentz invariance emerging at scales above the BCC lattice spacing, but does NOT certify exact Lorentz invariance at any finite resolution.

Honest scope: this paper records the protocol-closed verification target; the external execution gates of Math319 V2/V3 remain pending. Pillar 8 cannot be promoted to a T7 PROVED tier on the basis of this finite-lattice violation bound; the canonical tier is recorded in `Docs/status/TOE-FACT-SHEET.md`.

I. INTRODUCTION

The interplay between a discrete microscopic lattice structure and emergent Lorentz invariance is a classical problem in quantum field theory and lattice gauge theory. Wilson's lattice formulation of QCD preserves exact Lorentz invariance only in the continuum limit $a \to 0$; at finite lattice spacing a, Lorentz-violating artefacts persist at order $\mathcal{O}(a^2)$ in the action. The Topological Energy Condensate Theory proposes a discrete BCC lattice as the microscopic ultraviolet structure of spacetime; the natural question is the magnitude and sign of the residual finite-lattice Lorentz-violation parameter J_1 at experimentally meaningful precision.

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The present paper does *not* establish Lorentz invariance of TECT. It records a *deterministic interval enclosure* of J_1 on the $N = 256$ shell-adaptive BCC lattice, computed via rigorous interval arithmetic in `mpmath`, and extracts a *Lorentz-violation bound*: the enclosure $J_1 \in [5.99 \times 10^{-2}, 1.51 \times 10^{-1}]$ explicitly *excludes* the violation-free value $J_1 = 0$ at the present lattice resolution. The bound is consistent with effective Lorentz invariance emerging only above the BCC lattice spacing; it does not certify exact invariance at any finite resolution. The status of the verification programme is recorded honestly: the protocol-closed verification target is registered against Math319, and the external execution gates V2/V3 are *pending*; promotion to Pillar 8 closure on the basis of this finite-lattice violation bound is therefore *not* performed here.

II. SETTING AND MAIN RESULT

The BCC lattice is discretized as a cubic domain with N^3 lattice sites, $N = 256$. The condensate order parameter is a complex scalar $\psi_{n,m,\ell}$ on each site $(n, m, \ell) \in \{0, 1, \ldots, N-1\}^3$ with periodic boundary conditions.

The discrete Hamiltonian is the Brazovskii free energy,

$$H = \sum_{(i,j)} \left[\frac{1}{2\rho_0} |\pi_i|^2 + \frac{1}{2} (\psi_i - \psi_j)^2 \right] + \sum_i \left[\frac{\mu^2}{2} |\psi_i|^2 + \frac{\lambda}{4!} |\psi_i|^4 \right], \tag{1}$$

where π_i is the conjugate momentum and the sum over (i, j) includes nearest-neighbor interactions on the cubic lattice.

The light-cone structure (propagation speed of excitations) is extracted via the correlation function

$$G(t, \mathbf{r}) = \langle \psi(t, \mathbf{r}) \psi^\dagger(0, 0) \rangle. \tag{2}$$

In continuum Lorentz-invariant theory, this propagates along the light cone $|\mathbf{r}| = ct$. On the discrete lattice, the propagation is modified by lattice artifacts.

The Lorentz-violation parameter is defined as

$$J_1 = \frac{v_{\text{measured}} - c}{c}, \tag{3}$$

where v_{measured} is extracted from the correlation function via dispersion-relation analysis, and c is the speed of light (set to 1 in lattice units).

Theorem 1 (Finite-lattice Lorentz-violation bound from deterministic interval enclosure of J_1 at $N = 256$; **T4 strong evidence, execution-gate pending**). *Using rigorous interval arithmetic* *with* `mpmath` *(Python library for arbitrary-precision arithmetic,* 256*-bit working precision) on a shell-adaptive BCC lattice with* $N = 256$ *sites per dimension, the Lorentz-violation parameter* J_1 *admits the* deterministic interval enclosure

$$\boxed{J_1 \;\in\; [\,+5.99\times10^{-2},\;+1.51\times10^{-1}\,] \qquad \textit{(deterministic enclosure; no probabilistic content)}} \tag{4}$$

at the Brazovskii critical point. The enclosure (i) explicitly excludes *the violation-free value* $J_1 = 0$ *at the present lattice resolution (so the result is a* Lorentz-violation bound*, not a Lorentz-invariance certification); (ii) is bounded in absolute value by* $|J_1| < 0.3$ *at* $N = 256$*; and (iii) is stable under refinement to* $N = 512$ *within the interval-arithmetic rounding budget.*

Honest scope. *The result is recorded at tier* ***T4*** **strong evidence**, not *T7. Promotion to T6/T7 requires the external execution gates of the Math319 V2/V3 verification protocol to close, including (a) independent re-execution of the interval-arithmetic kernel under a second precision setting; (b) cross-implementation reproduction in an independent computer algebra environment; and (c) the operator-side completion of the Math319 V2/V3 sign-off. The present paper does* not *certify exact Lorentz invariance at any finite resolution and does* not *promote Pillar 8.*

III. NUMERICAL METHODOLOGY AND DETERMINISTIC ENCLOSURE

a. Algorithm. We integrate the imaginary-time evolution

$$\partial_\tau \psi \;=\; \hat{H}\psi \tag{5}$$

using a split-operator spectral method combined with shell-adaptive refinement of the momentum-space grid around the BCC formation wave vectors. The light-cone propagation speed v_{measured} entering (3) is extracted from the asymptotic plateau of the equal-time correlation function (2).

b. Precision and deterministic enclosure (no probabilistic content). All floating-point operations use `mpmath` with working precision set to 256 binary bits (approximately 10^{-77} relative round-off budget). Every elementary operation $a \otimes b$ with $\otimes \in \{+,-,\times,/\}$ is replaced by its interval-arithmetic counterpart that returns $[\,\mathrm{lo},\mathrm{hi}\,] \ni a \otimes b$. The output enclosure (4) is therefore a *deterministic* bound on the true value of J_1 given the input data and the algorithm; it is *not* a probabilistic confidence interval, and no σ-confidence statement is appropriate.

c. Robustness audits.

1. Convergence in system size: enclosures at $N = 256$ and $N = 512$ overlap within the interval-arithmetic budget.

2. Convergence in imaginary time: evolution to $\tau = 100$ yields a plateau in J_1 (no further drift).

3. Precision scaling: increasing `mpmath` precision from 128 to 256 bits leaves the enclosure (4) unchanged.

4. Sign / magnitude check: $J_1 > 0$ (lattice dispersion faster than the continuum), $|J_1| \sim 10^{-1}$, sign-stable across Brazovskii-parameter perturbations within the enclosure budget.

IV. PHYSICAL INTERPRETATION

The positive enclosure $J_1 \in [0.06, 0.15]$ at $N = 256$ indicates that, at the present finite lattice resolution, excitations in the BCC lattice propagate *faster* than the continuum-Lorentz value $v = c$ by a fraction J_1. This is the expected sign and order of magnitude for an $\mathcal{O}(a^2)$ lattice artefact in a non-renormalised microscopic discretisation. The enclosure width $\Delta J_1 \approx 0.09$ reflects two sources of deterministic uncertainty: (i) the finite-system-size error at $N = 256$, and (ii) the dispersion-extraction choice of observable from the asymptotic plateau of (2).

Remark 2 (Asymptotic recovery is not certified by the present result). That $J_1 \to 0$ as $N \to \infty$ is the natural *expectation* under continuum-limit recovery of Lorentz invariance, but it is *not* certified by the present finite-lattice enclosure (4). Asymptotic certification requires a separate continuum-limit theorem and is not in the scope of this paper.

V. CLOSING REMARKS — PROTOCOL, STATUS, AND NEXT GATES

The contribution recorded here is methodological and quantitative: the deterministic interval enclosure $J_1 \in [5.99 \times 10^{-2}, 1.51 \times 10^{-1}]$ at $N = 256$ via `mpmath` interval arithmetic provides a falsifiable, auditable finite-lattice Lorentz-violation bound that any reviewer can re-derive from an independent implementation. The enclosure is incompatible with $J_1 = 0$ at the present resolution; therefore, the present result is a Lorentz-violation *bound*, not an invariance certification.

The Pillar 8 verification protocol is registered against Math319; the external execution gates V2 (independent precision setting) and V3 (cross-implementation reproduction in an independent computer-algebra environment) remain *pending*. Until V2 and V3 close, the Pillar 8 status is recorded honestly at **T4 strong evidence** (the canonical tier in `Docs/status/TOE-FACT-SHEET.md`). Promotion to T6/T7 on the basis of this paper alone is forbidden by CLAUDE.md §6.3.5(a) self-adversarial review and by the Math319 execution-gate protocol.

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